

Master Execution Roadmap

Master Execution Roadmap

Master Execution Roadmap

Project: VelaDial

Status: Active

Purpose: This document serves as the real execution contract for the VelaDial project. It maps out the entire remaining build sequence, defining all validation gates, decision points, phase dependencies, and strict definitions of done.

SECTION A — Current Project State

See also: [docs/FINAL_COMPLETION_ROADMAP.md](#) — the concrete punch list of what remains before VelaDial v1 is ready for hardware validation, what only Hardik can finish with physical hardware, the recommended near-term PR sequence (A–E), and the explicit no clean-repo migration yet decision.

The project has established a solid documentation foundation, but firmware and hardware validation remain in preliminary states.

Pull requests **#1 through #35** have been merged. PRs **#18–#24** added the Master Execution Roadmap, door-side firmware draft, bedside firmware draft, Raspberry Pi / Home Assistant setup guide, full E2E setup and validation guide, UI/UX guide integration plan, and Claude review package. PRs **#25–#27** addressed the Claude review findings: immediate cleanup (PR **#25**), VL53L4CD decision recording (PR **#26**), and pre-migration repo cleanup (PR **#27**). PR **#29** added the Final Completion Roadmap. PR **#30** added ESPHome compile CI and secrets template hardening (and reverted the PR **#25** preset value type regression). PR **#31** enabled door brightness arc direct-touch with CI validation. PR **#32** added Home Assistant config examples. PR **#34** added the premium UI concept direction package (20 concepts, 6 documents), superseding Claude's PR **#33**. PR **#35** added the UI concept prototype framework.

The **independent Claude review** of PRs #15–#24 returned a verdict of **REQUEST CHANGES** with three blocking items. All three have been resolved: Step 15B gate waiver noted (PR #25), VL53L4CD decision recorded as Option B / defer to v2 (PR #26), and `.gitignore` hardened with `secrets.yaml` patterns (PR #25).

Note on PR #25's preset value type change: PR #25 converted the preset

`homeassistant.action` `data:` values (`color_temp` , `brightness_pct`) from quoted strings to YAML numerics. CI introduced in PR #30 surfaced that ESPHome's schema rejects this — `data:` values must be strings. PR #30 reverts those 8 values back to quoted strings. The lesson: ESPHome `homeassistant.action` `data:` literals must be strings (or `!lambda` expressions). Future PRs should not "clean up" these to numerics.

While the ELECROW GPIOs are source-confirmed, physical hardware validation remains strictly **NOT TESTED**. The current YAML files (`esphome/door_side_rotary.yaml` and `esphome/bedside_gesture.yaml`) are production-oriented **drafts** — compile-time reviewed only — and are explicitly labeled `DRAFT – COMPILER PASSED / HARDWARE VALIDATION PENDING` in their headers. No physical PASS results currently exist.

SECTION B — Locked v1 Scope

The following behaviors and constraints are strictly locked for the version 1 (v1) build.

Door-side Controller

The door-side UI consists of exactly three pages: Power, Brightness, and Presets. There is no required Environment page, and the SHT45 sensor is reserved for secondary or future diagnostics only. The Presets page features exactly four locked presets: Warm White, Soft Amber, Neutral White, and Low Nightlight.

The device employs a strict wake-only-first behavior. Touching the screen, rotating the knob, or pressing the knob while the device is asleep will only wake the display. A second deliberate action while the display is awake is required to send a command or toggle a state.

Page navigation is handled via horizontal swipes: a left swipe moves to the next page, and a right swipe moves to the previous page. A 3-dot indicator displays the current page, with the active dot colored amber.

Knob press behavior varies by page: - **Power page:** Toggles the bedroom light group. - **Brightness page:** Returns to the Power page. - **Presets page:** Applies the currently highlighted preset. - **While asleep:** Wakes the display only.

Bedside Controller

The bedside controller utilizes APDS-9960 standalone gestures for v1. A left gesture turns the bedroom group off, while a right gesture turns the bedroom group on. A cooldown period prevents repeated triggers.

VL53L4CD v1 decision — Option B (deferred), recorded 2026-05-25 by Hardik. The VL53L4CD standalone hand-hold nightlight is **NOT** part of v1. The sensor remains physically present on the bedside breadboard for v2 / future use, but no v1 firmware, ESPHome YAML, Home Assistant logic, or UI element references it. APDS-9960 / VL53L4CD sensor fusion is independently reserved for v2 or future builds; there is absolutely no sensor fusion in v1. VL53LOX fallback (Option C) is **not** approved; do not introduce it without an explicit Hardik decision.

System Path

The control path flows from the ESPHome devices to Home Assistant running on a Raspberry Pi, then via LocalTuya over the local LAN to the `light.bedroom_group`.

SECTION C — Source Confirmed vs Physical Test Pending

 **WARNING:** Source-confirmed does not equal physically validated.

ITEM	SOURCE-CONFIRMED STATUS	PHYSICAL STATUS	EVIDENCE FILE	NEXT ACTION
ELECROW GPIOs	Confirmed	NOT TESTED	<code>docs/validation/elecrow_source_confirmation_matrix.md</code>	Awaiting Hardik physical validation
Physical board revision	Pending	NOT TESTED	<code>hardware/validation_results.md</code>	Photograph silkscreen/sticker
ESPHome compile	Pending	NOT TESTED	<code>hardware/validation_results.md</code>	Run compile command
Flash	Pending	NOT TESTED	<code>hardware/validation_results.md</code>	Flash device and capture logs
Serial logs	Pending		<code>hardware/validation_results.md</code>	

ITEM	SOURCE- CONFIRMED STATUS	PHYSICAL STATUS	EVIDENCE FILE	NEXT ACTION
		NOT TESTED		Capture clean boot logs
I2C scan	Pending	NOT TESTED	hardware/validation_results.md	Run I2C scan on both buses
Display test	Pending	NOT TESTED	hardware/validation_results.md	Verify GC9A01A output
Touch test	Pending	NOT TESTED	hardware/validation_results.md	Verify CST816 coordinates
Encoder test	Pending	NOT TESTED	hardware/validation_results.md	Verify CW/ CCW and press
WS2812 test	Pending	NOT TESTED	hardware/validation_results.md	Verify 5 LEDs
30-minute soak	Pending	NOT TESTED	hardware/validation_results.md	Run stability test

SECTION D — Firmware Execution Plan

This section outlines the plan for firmware development across the three target platforms. No firmware is written in this phase.

Target 1: Door-side ELECROW ESP32-S3

- **Intended YAML file path:** [esphome/door_side_rotary.yaml](#)
- **Intended entities:** Display, touch, rotary encoder, backlight PWM, WS2812 LEDs, TSL2591, SHT45.
- **Display/LVGL features:** 3-page UI (Power, Brightness, Presets), 3-dot indicator, amber accents.
- **Wake-only-first behavior plan:** Implement logic to intercept first input when asleep.
- **Power/brightness/preset behavior:** Map knob and touch inputs to specific HA actions per page.

- **Backlight behavior:** TSL2591 lux drives PWM locally.
- **Secrets handling:** Use `!secret` for WiFi, API, and OTA passwords.
- **Compile command:** `esphome compile esphome/door_side_rotary.yaml`
- **Flash command:** `esphome run esphome/door_side_rotary.yaml`
- **Validation checklist:** Display renders, touch accurate, encoder responsive, backlight adapts.
- **Rollback plan:** Revert to bring-up YAML.
- **Untested items:** All physical hardware interactions.

Target 2: Bedside ESP32-C6 Feather

- **Intended YAML file path:** `esphome/bedside_gesture.yaml`
- **APDS-9960 gesture logic:** Standalone left/right detection.
- **Left/right actions:** Left = off, Right = on.
- **Cooldown behavior:** ~1s delay after successful gesture.
- **Home Assistant target entity:** `light.bedroom_group`
- **Secrets handling:** Use `!secret` for WiFi, API, and OTA passwords.
- **Compile command:** `esphome compile esphome/bedside_gesture.yaml`
- **Flash command:** `esphome run esphome/bedside_gesture.yaml`
- **Validation checklist:** Gestures trigger HA actions, cooldown prevents spam.
- **Rollback plan:** Revert to bring-up YAML.
- **Untested items:** All physical hardware interactions.
- **Explicitly no sensor fusion:** APDS-9960 operates entirely independently of VL53L4CD.

Target 3: Raspberry Pi / Home Assistant

- **Home Assistant role:** Central hub for state management and automation.
 - **ESPHome add-on role:** Compiles and manages ESP32 nodes.
 - **LocalTuya role:** Provides local LAN control of Tuya bulbs.
 - `light.bedroom_group`: The single target entity for all lighting commands.
 - **Network assumptions:** Local LAN must be stable; internet not required for normal control.
 - **Static IP/reservation recommendation:** Assign static IPs to Pi, ESP32s, and Tuya bulbs.
 - **secrets.yaml handling:** Store all sensitive keys locally on the Pi.
 - **Backup/restore:** Configure automated HA backups.
 - **Troubleshooting/log collection:** Use HA log viewer and ESPHome serial logs.
-

SECTION E — UI/UX Guide Integration Plan

Pending Hardik UI/UX Guide Integration

Hardik possesses UI/UX guidance that may not yet be present in the GitHub repository.

- **What Hardik must provide:** The detailed UI/UX guide or specific design assets.
- **How it will be integrated:** The guide will be reviewed and its styling/animation directives applied to the LVGL implementation.
- **Rule:** The UI guide cannot add pages unless Hardik explicitly approves.
- **Locked v1:** The interface remains exactly 3 pages.
- **No Environment page:** This remains excluded unless approved for a future v2.
- **Premium UI:** Premium UI directives can affect styling and animation, but cannot alter the core v1 scope unless explicitly approved.

SECTION F — Corrected Phase Plan

The execution must strictly follow this sequence.

PHASE	OUTPUT	DETAILS
Phase 0 — Master Execution Roadmap	docs/MASTER_EXECUTION_ROADMAP.md	Current task. PR required.
Phase 1 — Door-side ESP32-S3 firmware draft	Draft update to esphome/door_side_rotary.yaml	IN REVIEW — draft created; compile reported successful; physical validation pending; PR pending.
Phase 2 — Bedside ESP32-C6 APDS firmware draft	Draft update to esphome/bedside_gesture.yaml	APDS left/right gestures, cooldown, no sensor fusion, no VL53L4CD unless separately approved.
Phase 3 — Raspberry Pi / Home	docs/setup/raspberry_pi_home_assistant_setup.md	Document HA, ESPHome, and LocalTuya configuration.

PHASE	OUTPUT	DETAILS
Assistant setup guide		
Phase 4 — Full E2E setup and validation guide	docs/setup/full_e2e_setup_and_validation_guide.md	Comprehensive testing plan.
Phase 5 — UI/UX guide integration plan	docs/ui/ux_guide_integration_plan.md	Integrate Hardik-provided UI guide if available.
Phase 6 — Claude review package	docs/review/claude_review_package.md	Prepare documentation for external review.
Phase 7 — VL53L4CD path	RESOLVED 2026-05-25 — Option B (defer to v2)	No v1 firmware/YAML/HA touches VL53L4CD. Re-opening the decision (for a v2 build) requires a new owner-recorded Status: block in docs/vl53l4cd_support_verification.md and a follow-up PR series updating PRD/Backend Schema/this roadmap.

SECTION G — Hardware E2E Test Backlog

The following tests must be completed and recorded as PASS before the system is considered validated.

- ELECROW board revision photo
- Front/back board photos
- Compile result
- Flash result
- Serial boot logs
- I2C scan

- GC9A01A display test
- Backlight PWM test
- CST816 touch address test
- CST816 touch coordinate/orientation test
- Rotary CW/CCW test
- Rotary press test
- WS2812 5 LED test
- 30-minute stability soak
- ESP32-C6 compile
- ESP32-C6 flash
- APDS-9960 I2C scan
- APDS left gesture test
- APDS right gesture test
- Gesture cooldown test
- Home Assistant service call test
- LocalTuya command path test
- `light.bedroom_group` on/off test
- Brightness test
- Preset test
- VL53L4CD support-gated test (only if approved)

SECTION H — Objective Completion Matrix

OBJECTIVE	STATUS	EVIDENCE	FILE / PR	NEXT ACTION
Roadmap	DONE	This document	<code>docs/MASTER_EXECUTION_ROADMAP.md</code>	PR #18 merged
Door firmware draft	DONE	PR #19 merged	<code>esphome/door_side_rotary.yaml</code>	compile PASSED; physical validation pending
Bedside firmware draft	DONE	PR #20 merged	<code>esphome/bedside_gesture.yaml</code>	compile PASSED; physical validation pending

OBJECTIVE	STATUS	EVIDENCE	FILE / PR	NEXT ACTION
				validation pending
Raspberry Pi guide	DONE	PR #21 merged	docs/setup/raspberry_pi_home_assistant_setup.md	HA/ESPHome LocalTuya NOT TESTED
E2E guide	DONE	PR #22 merged	docs/setup/full_e2e_setup_and_validation_guide.md	E2E validation NOT TESTED
UI/UX guide integration	DONE	PR #23 merged	docs/ui/ux_guide_integration_plan.md	Hardik UI guide PENDING
Claude review package	DONE	PR #24 merged	docs/review/claude_review_package.md	Claude review completed
UI concept direction package	DONE	PR #34 merged	docs/ui/ui_concept_direction_matrix.md , docs/ui/ui_concept_shortlist.md , docs/ui/ui_concept_selection_criteria.md , docs/ui/ui_concept_notes.md , docs/ui/ui_visual_batch_plan.md , docs/ui/ui_implementation_candidates.md	Complete
UI concept prototype framework	DONE	PR #35 merged	docs/ui/concepts/README.md , esphome/concepts/README.md	Complete
UI concept prototypes (1-20)	DONE	All 20 concepts complete (PRs #36-#55 merged)	docs/ui/concepts/ , esphome/concepts/	All 20 concept prototype delivered merged
Concept 01: Minimal Thermostat	MERGED	PR #36 merged	docs/ui/concepts/01_minimal_thermostat/ , esphome/concepts/door_side_concept_01_minimal_thermostat.yaml	Compile PASSED; merged
Concept 02: SmartKnob-Inspired Arc	MERGED	PR #37 merged	docs/ui/concepts/02_smartknob_arc/ , esphome/concepts/door_side_concept_02_smartknob_arc.yaml	Compile PASSED; merged
Concept 03: Large Center Power Button	MERGED	PR #38 merged	docs/ui/concepts/03_large_center_power_button/ , esphome/concepts/door_side_concept_03_large_center_power.yaml	Compile PASSED; merged

OBJECTIVE	STATUS	EVIDENCE	FILE / PR	NEXT ACTION
Concept 04: Single-Page Simple Mode	MERGED	PR #39 merged	docs/ui/concepts/04_single_page_simple_mode/ , esphome/concepts/ door_side_concept_04_single_page_simple.yaml	Compile PASSED; Gate G1;
Concept 05: Preset Ring UI	MERGED	PR #40 merged	docs/ui/concepts/05_preset_ring_ui/ , esphome/ concepts/door_side_concept_05_preset_ring.yaml	Compile PASSED; merged
Concept 06: Night Mode Ultra-Minimal	MERGED	PR #41 merged	docs/ui/concepts/06_night_mode_ultra_minimal/ , esphome/concepts/ door_side_concept_06_night_mode_ultra_minimal.yaml	Compile PASSED; concept; merged
Concept 07: Text-First Utility	MERGED	PR #42 merged	docs/ui/concepts/07_text_first_utility/ , esphome/ concepts/ door_side_concept_07_text_first_utility.yaml	Compile PASSED; candidate merged
Concept 08: Apple Watch Complications	MERGED	PR #43 merged	docs/ui/concepts/08_apple_watch_complications/ , esphome/concepts/ door_side_concept_08_apple_watch_complications.yaml	Compile PASSED; RECOMM for v1; v1- expanded adaptation merged
Concept 09: LED-Ring Status-First	MERGED	PR #44 merged	docs/ui/concepts/09_led_ring_status_first/ , esphome/concepts/ door_side_concept_09_led_ring_status_first.yaml	Compile PASSED; concept; ring as pr feedback merged
Concept 10: Three-Screen Tab Carousel	MERGED	PR #45 merged	docs/ui/concepts/10_three_screen_tab_carousel/ , esphome/concepts/ door_side_concept_10_three_screen_tab_carousel.yaml	Compile PASSED; v1 spec implemen merged
Concept 11: Brightness- First UI	MERGED	PR #46 merged	docs/ui/concepts/11_brightness_first_ui/ , esphome/ concepts/ door_side_concept_11_brightness_first_ui.yaml	Compile PASSED; innovation

OBJECTIVE	STATUS	EVIDENCE	FILE / PR	NEXT ACTION
				(brightness default); n
Concept 12: Door Switch Replacement	MERGED	PR #47 merged	<code>docs/ui/concepts/12_door_switch_replacement/</code> , <code>esphome/concepts/</code> <code>door_side_concept_12_door_switch_replacement.yaml</code>	Compile PASSED; screen po hero; gue friendly; n
Concept 13: Lunar Phase Visualization	MERGED	PR #48 merged	<code>docs/ui/concepts/13_lunar_phase_visualization/</code> , <code>esphome/concepts/</code> <code>door_side_concept_13_lunar_phase_visualization.yaml</code>	Compile PASSED; phase metaphor poetic/pre merged
Concept 14: Sundial Shadow UI	MERGED	PR #49 merged	<code>docs/ui/concepts/14_sundial_shadow_ui/</code> , <code>esphome/</code> <code>concepts/</code> <code>door_side_concept_14_sundial_shadow_ui.yaml</code>	Compile PASSED; shadow metaphor architectu premium; merged
Concept 15: Tree Ring Growth Pattern	MERGED	PR #50 merged	<code>docs/ui/concepts/15_tree_ring_growth_pattern/</code> , <code>esphome/concepts/</code> <code>door_side_concept_15_tree_ring_growth_pattern.yaml</code>	Compile PASSED; ring meta organic/ premium; merged
Concept 16: Topographic Contour Map	MERGED	PR #51 merged	<code>docs/ui/concepts/16_topographic_contour_map/</code> , <code>esphome/concepts/</code> <code>door_side_concept_16_topographic_contour_map.yaml</code>	Compile PASSED; Topograp contour metaphor cartograp premium; merged
Concept 17: Iris Aperture Mechanism	MERGED	PR #52 merged	<code>docs/ui/concepts/17_iris_aperture_mechanism/</code> , <code>esphome/concepts/</code> <code>door_side_concept_17_iris_aperture_mechanism.yaml</code>	Compile PASSED; aperture metaphor

OBJECTIVE	STATUS	EVIDENCE	FILE / PR	NEXT ACTION
				optical/ mechanic premium; merged
Concept 18: Radar Sweep Animation	MERGED	PR #53 merged	<code>docs/ui/concepts/18_radar_sweep_animation/</code> , <code>esphome/concepts/</code> <code>door_side_concept_18_radar_sweep_animation.yaml</code>	Compile PASSED; sweep metaphor futuristic/ scanning, premium; merged
Concept 19: Vinyl DJ Crossfader	MERGED	PR #54 merged	<code>docs/ui/concepts/19_vinyl_dj_crossfader/</code> , <code>esphome/</code> <code>concepts/</code> <code>door_side_concept_19_vinyl_dj_crossfader.yaml</code>	Compile PASSED; turntable metaphor audiophile premium; merged
Concept 20: Eclipse Corona	MERGED	PR #55 merged	<code>docs/ui/concepts/20_eclipse_corona/</code> , <code>esphome/</code> <code>concepts/door_side_concept_20_eclipse_corona.yaml</code>	Compile PASSED; corona metaphor cinematic premium/ FINAL CONCEPT merged
Multi-Theme Firmware Engine (candidate)	MERGED	PR #56 merged	<code>esphome/door_side_rotary.yaml</code> , <code>docs/ui/</code> <code>theme_visual_upgrade_plan.md</code> , <code>docs/ui/</code> <code>multi_theme_firmware_engine.md</code> , <code>docs/ui/</code> <code>theme_selector_ux.md</code>	Compile- passing m theme fir candidate selectable themes in binary, th selector v long-pres shared co logic, per

OBJECTIVE	STATUS	EVIDENCE	FILE / PR	NEXT ACTION
				theme sel theme-co LED ring. Compile PASSED. Hardware validation pending. visual refinemen follow har testing.
Multi-Theme Hardware Validation Package	IN REVIEW	PR pending	docs/validation/multi_theme_hardware_validation_plan.md , docs/validation/multi_theme_test_results_template.md	Document checklist package t Hardik to physically validate th multi-ther firmware. results ma PENDING TESTED u Hardik pe testing.
Claude independent review	COMPLETED — REQUEST CHANGES	Review delivered to Hardik	n/a	Findings t via cleanu series
Immediate cleanup PR	DONE	PR #25 merged	.gitignore , hardware/validation_results.md , esphome/door_side_rotary.yaml , esphome/bedside_gesture.yaml , docs/MASTER_EXECUTION_ROADMAP.md	Complete
VL53L4CD decision recording	DONE	PR #26 merged	docs/vl53l4cd_support_verification.md , docs/01_PRD.md , docs/05_Backend_Schema.md	Option B recorded
Pre-migration cleanup PR	DONE	PR #27 merged	Repo-wide cleanup	Complete

OBJECTIVE	STATUS	EVIDENCE	FILE / PR	NEXT ACTION
Final Completion Roadmap	DONE	PR #29 merged	docs/FINAL_COMPLETION_ROADMAP.md	Complete
Compile CI + secrets hardening	DONE	PR #30 merged	.github/workflows/ , esphome/secrets.yaml.example	CI active
Door brightness arc direct-touch	DONE	PR #31 merged	esphome/door_side_rotary.yaml	CI validate
HA config examples	DONE	PR #32 merged	setup/	Complete
ELECROW physical validation	HARDWARE TEST PENDING	None	hardware/validation_results.md	Hardik to
Bedside APDS validation	HARDWARE TEST PENDING	None	hardware/validation_results.md	Hardik to
HA command-path validation	HARDWARE TEST PENDING	None	hardware/validation_results.md	Hardik to
VL53L4CD decision	DEFERRED v1 — Option B	Owner-recorded 2026-05-25	docs/vl53l4cd_support_verification.md , docs/01_PRD.md , docs/05_Backend_Schema.md	v1 ships A only beds revisit for
No sensor fusion compliance	IN PROGRESS	Roadmap constraints	Multiple	Maintain constraints
Secrets safety	DONE	.gitignore hardened in PR #25	.gitignore , setup guide	Maintain constraints

SECTION I — Definition of Done

The project is not considered "done" until all of the following conditions are met:

- The roadmap exists and is approved.
 - The door firmware draft exists.
 - The bedside firmware draft exists.
 - The Raspberry Pi/Home Assistant setup guide exists.
 - The full E2E setup/validation guide exists.
 - The UI/UX guide is integrated or clearly marked as pending.
 - The Claude review package exists.
 - All hardware tests are listed in the backlog.
 - All untested items are explicitly marked as NOT TESTED or HARDWARE TEST PENDING.
 - No sensor fusion has been added to the v1 implementation.
 - No unsupported VL53L4CD claims are made.
 - No secrets or credentials are committed to the repository.
 - All branches are pushed to the remote repository.
 - All Pull Requests are created.
 - Hardik manually reviews and merges all Pull Requests.
-

SECTION J — GitHub / PR Requirement

- If GitHub access works: Push the branch and open a Pull Request.
- If GitHub access does not work: Mark GitHub PR creation as BLOCKED and provide the exact error.
- Do not substitute a zip file for a PR without explicitly marking PR creation as BLOCKED.